



TOPIC 4: GRAPHICS IN R FOR CATEGORICAL DATA

USING R TO GRAPHICALLY DISPLAY CATEGORICAL/ QUALITATIVE DATA IN R

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Objectives: Student will be able to do the following Using R:

- For Categorical Data:
 - Frequency Tables
 - Pie Charts
 - Bar Graphs

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Frequency Table for Categorical DATA

- It is always a good idea to let R recognize a variable as qualitative/categorical. Below is the code:

```
>#This code makes R recognize the variable Blood_Type as Qualitative/Categorical  
>Blood$Blood_Type=as.factor(Blood$Blood_Type)
```

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Frequency Table for Categorical DATA

- Creating the frequency table

```
>#Getting a Frequency Table  
>t=table(Blood)  
>#To see the Frequency Table  
>t
```

- Output:

```
Blood  
 A AB  B  O  
15  4  6 25
```

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Frequency Table for Categorical DATA

- Creating the frequency table with proportions

```
>#Getting a Frequency Table with proportions after you have created the frequency table  
>t_proportions=t/sum(t)  
>#To see the Frequency Table with proportions  
>t_proportions
```

- Output:

```
Blood  
  A  AB  B  O  
0.30 0.08 0.12 0.50
```

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Frequency Table for Categorical DATA

- Creating the frequency table with percentages

```
>#First, I must take data and create percentages after I have created my frequency table  
>t_percent=100*t/sum(t)  
>#To see the Relative Frequency Table  
>t_percent  
> #to create the table to include percent symbol for clarity, format percentages  
>t_percent2<-paste0(round(t_percent,2), "%")  
>#print table with variable name  
>names(t_percent2)<-names(t_percent)  
>t_percent2
```

- Output:

```
      A  AB   B   O  
"30%" "8%" "12%" "50%"
```

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Pie Chart for Categorical DATA

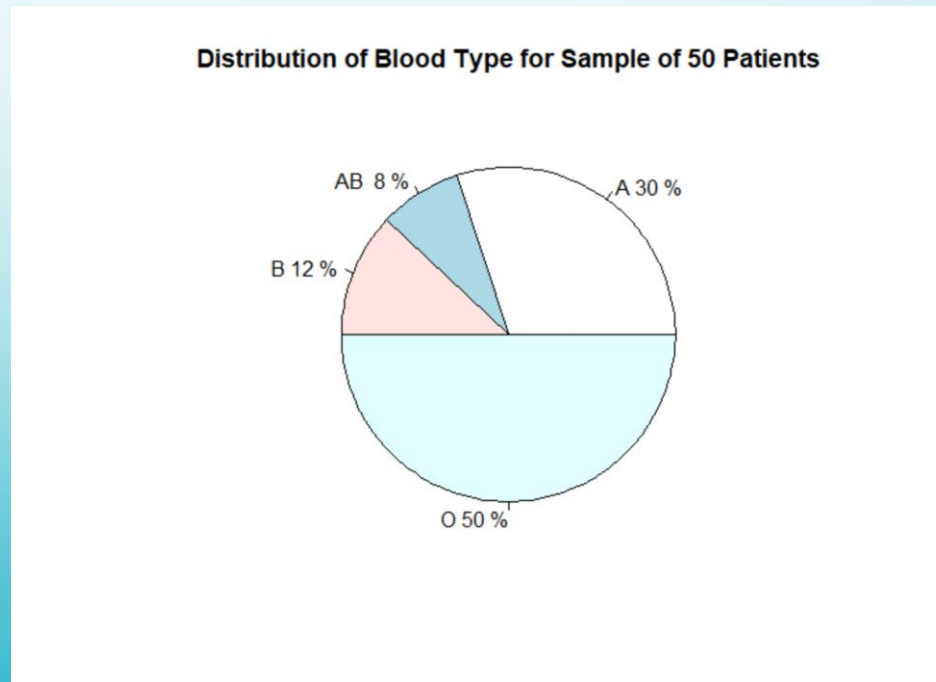
```
>#This code makes R recognize the variable Blood_Type as Qualitative/Categorical  
>Blood$Blood_Type=as.factor(Blood$Blood_Type)  
>#Getting a Frequency Table  
>t=table(Blood)  
>#First I must take data and create percentages  
>percent=100*t/sum(t)  
>#Creating a Pie Chart with slice labels and percentages  
>format_with_percent=format(percent,digits=3)  
>lbl=paste(c("A","AB","B","O"), format_with_percent,"%",sep=" ")  
>pie(percent, label = lbl,main = "Distribution of Blood Type for Sample of 50 Patients")
```

Results on next slide

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Pie Chart for Categorical DATA

You must click “Plots” on the bottom right window, click “Export”, click “Copy to Clipboard...”, in pop up window click “Copy Plot” at bottom right, then paste into documents. It will look like image below:

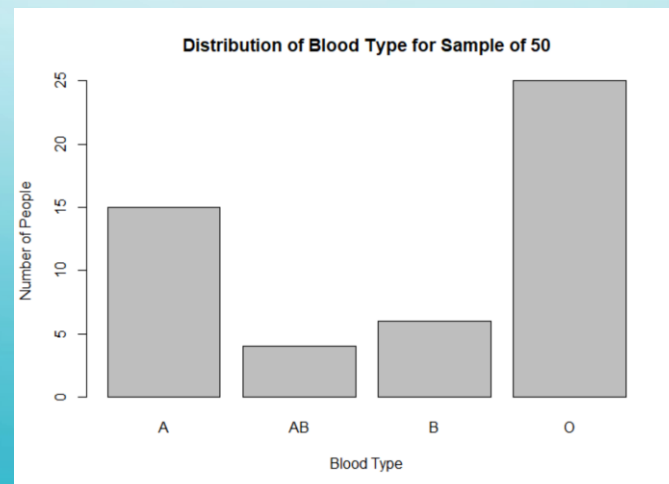


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Frequency Bar Graph for Categorical DATA

```
>>#This code makes R recognize the variable Blood_Type as Qualitative/Categorical  
>Blood$Blood_Type=as.factor(Blood$Blood_Type)  
>#Getting a Frequency Table  
>t=table(Blood)  
>#Creating a Frequency Bar Graph with Main Title and x and y labels  
>barplot(t,main = "Distribution of Blood Type for Sample of 50",xlab = "Blood Type",ylab = "Number of  
People")
```

Result:

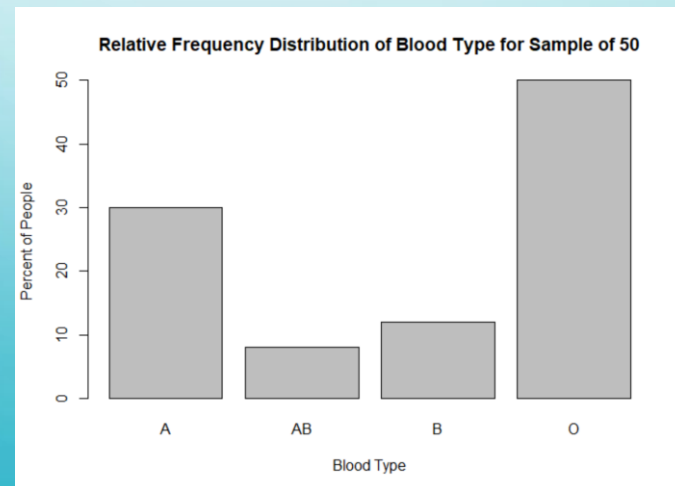


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Frequency Bar Graph for Categorical DATA

```
>#This code makes R recognize the variable Blood_Type as Qualitative/Categorical  
>Blood$Blood_Type=as.factor(Blood$Blood_Type)  
>#Getting a Frequency Table  
>t=table(Blood)  
>#Creating a Relative Frequency Bar Graph with Main Title and x and y labels  
>barplot(percent,main = "Relative Frequency Distribution of Blood Type for Sample of 50",xlab = "Blood  
Type",ylab = "Percent of People")
```

Result:



TOPIC 4: GRAPHICS IN R

Wrap – Up

- For Categorical Data:
 - Frequency Tables
 - Pie Graphs
 - Bar Graphs